

Complimentary Function (C.F.)

The C.F. is the solution of the d. eqⁿ.

$$f(D).y = 0 \quad \text{--- (3)}$$

To find the solution of eqⁿ. (3), we write an equation called auxiliary equation.

$$\text{i.e. } f(m) = 0$$

$$\Rightarrow a_0 m^n + a_1 m^{n-1} + \dots + a_n = 0 \quad \text{--- (4)}$$

Let, the roots of this eqⁿ. be $m_1, m_2, \dots, m_n$.

Case - I: If roots are real and distinct.

$$\text{C.F.} = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

Case - II: If roots are real and equal.

$$\text{i.e. } m_1 = m_2 = m_3 = \dots = m_n = m \text{ (say)}$$

$$\text{C.F.} = [C_1 + C_2 x + C_3 x^2 + \dots + C_n x^{n-1}] e^{mx}$$

Case - III: If roots are complex and distinct.

$$\text{i.e. } m = \alpha \pm i\beta, \gamma \pm i\delta$$

$$\begin{aligned} \text{C.F.} = & e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x] \\ & + e^{\gamma x} [C_3 \cos \delta x + C_4 \sin \delta x] \end{aligned}$$

Case - V: If roots are irrational i.e. $m = \alpha \pm \sqrt{\beta}$
C.F. = $e^{\alpha x} [C_1 \cosh(\beta x) + C_2 \sinh(\beta x)]$ Page _____ Date _____

Case - IV: When roots are complex and equal.
i.e. $m = \alpha \pm i\beta, \alpha \pm i\beta$

$$\text{C.F.} = e^{\alpha x} [(C_1 + C_2 x) \cos \beta x + (C_3 + C_4 x) \sin \beta x]$$

→ For two equal and real roots,

$$\text{C.F.} = (C_1 + C_2 x) e^{mx}$$

→ For three equal and real roots,

$$\text{C.F.} = (C_1 + C_2 x + C_3 x^2) e^{mx}$$

Q) Solve:

$$(i) \frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 15y = 0$$

Solⁿ:

$$D = \frac{d}{dx}$$

So, the given eqⁿ. can be written as,

$$[D^2 - 8D + 15]y = 0$$

The auxiliary eqⁿ. is,

$$m^2 - 8m + 15 = 0$$

$$\text{or } m^2 - 5m - 3m + 15 = 0$$

$$\text{or } m(m-5) - 3(m-5) = 0$$

$$\text{or } (m-5)(m-3) = 0$$

$$\Rightarrow m = 3, 5$$

Here, the roots are real and distinct,

$$\therefore \text{C.F.} = C_1 e^{3x} + C_2 e^{5x}$$

Since, $Q = 0$, the general solution is,

$$y = \text{C.F.} = C_1 e^{3x} + C_2 e^{5x} \quad \underline{\text{Ans}}$$

$$(ii) \frac{d^2 y}{dx^2} + y = 0$$

Solⁿ:

$$(D^2 + 1)y = 0$$

The auxiliary eqⁿ. is,

$$m^2 + 1 = 0$$

$$\text{or, } m^2 = -1$$

$$\text{or, } m^2 = i^2$$

$$\text{or, } m = \pm i$$

$$\Rightarrow m = 0 \pm i.1$$

$$\therefore \text{C.F.} = e^{0x} [C_1 \cos x + C_2 \sin x]$$

$$\therefore y = \text{C.F.} = C_1 \cos x + C_2 \sin x \quad \underline{\text{Ans}}$$

(iii) $[D^4 + 8D^2 + 16]y = 0$

Solⁿ:

The auxiliary eqⁿ is,

$$m^4 + 8m^2 + 16 = 0$$

$$\text{or, } (m^2 + 4)^2 = 0$$

$$\text{or, } (m^2 + 4)(m^2 + 4) = 0$$

$$\Rightarrow m^2 = -4, -4$$

$$\Rightarrow m^2 = 4i^2, 4i^2$$

$$\Rightarrow m = \pm 2i, \pm 2i$$

$$\therefore \text{C.F.} = e^{0x} [(C_1 + C_2 x) \cos 2x + (C_3 + C_4 x) \sin 2x]$$

$$\therefore y = \text{C.F.} = (C_1 + C_2 x) \cos 2x + (C_3 + C_4 x) \sin 2x$$

(iv) $\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 3y = 0$

Solⁿ:

$$[D^2 + D - 3]y = 0$$

The auxiliary eqⁿ is,

$$m^2 + m - 3 = 0$$

$$[ax^2 + bx + c = 0]$$

$$\Rightarrow m = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-3)}}{2 \times 1}$$

$$\left[x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$\Rightarrow m = \frac{-1 \pm \sqrt{13}}{2}$$

$$\Rightarrow m = \frac{-1 + \sqrt{13}}{2}, \frac{-1 - \sqrt{13}}{2}$$

$$\therefore \text{C.F.} = y = C_1 e^{\left(\frac{-1 + \sqrt{13}}{2}\right)x} + C_2 e^{\left(\frac{-1 - \sqrt{13}}{2}\right)x} \quad \underline{\text{Ans}}$$

$$(iv) \frac{d^4 y}{dx^4} + 32 \frac{d^2 y}{dx^2} + 256 y = 0$$

Solⁿ:

$$[D^4 + 32D^2 + 256] y = 0$$

The auxiliary eqⁿ. is,

$$m^4 + 32m^2 + 256 = 0$$

$$\Rightarrow (m^2 + 16)^2 = 0$$

$$\Rightarrow m^2 = -16, -16$$

$$\Rightarrow m^2 = 16i^2, 16i^2$$

$$\Rightarrow m = \pm 4i, \pm 4i$$

$$\therefore \text{C.F.} = e^{0x} [(C_1 + C_2 x) \cos 4x + (C_3 + C_4 x) \sin 4x]$$

$$\therefore y = \text{C.F.} = (C_1 + C_2 x) \cos 4x + (C_3 + C_4 x) \sin 4x \quad \underline{\text{Ans}}$$

(vi) $\frac{d^2y}{dx^2} + u^2y = 0$

Solⁿ:

$$[D^2 + u^2]y = 0$$

The auxiliary eqⁿ is,

$$m^2 + u^2 = 0$$

$$\Rightarrow m^2 = -u^2$$

$$\Rightarrow m^2 = u^2 i^2$$

$$\Rightarrow m = \pm ui$$

$$\therefore C.F. = e^{ax} (C_1 \cos ux + C_2 \sin ux)$$

$$\therefore y = C.F. = C_1 \cos ux + C_2 \sin ux \quad \underline{\text{Ans}}$$

(vii) $(D^2 + 1)^3 y = 0$

Solⁿ:

The auxiliary eqⁿ is,

$$(m^2 + 1)^3 = 0$$

$$\Rightarrow m^6 + 3m^4 + 3m^2 + 1 = 0$$

$$\Rightarrow (m^2 + 1)(m^2 + 1)(m^2 + 1) = 0$$

$$\Rightarrow m^2 = -1, -1, -1$$

$$\Rightarrow m^2 = i^2, i^2, i^2$$

$$\Rightarrow m = \pm i, \pm i, \pm i$$

$$\therefore C.F. = e^{0.x} [(C_1 + C_2x + C_3x^2) \cos x + (C_4 + C_5x + C_6x^2) \sin x]$$

$$\therefore y = C.F. = (C_1 + C_2x + C_3x^2) \cos x + (C_4 + C_5x + C_6x^2) \sin x \quad \underline{\text{Ans}}$$

(viii) $(D^5 - D^3)y = 0$

Solⁿ:

The A.E. is,

$$m^5 - m^3 = 0$$

$$\Rightarrow m^3(m^2 - 1) = 0$$

$$\Rightarrow m^3 = 0$$

$$\Rightarrow m = 0, 0, 0, \quad m^2 - 1 = 0$$

$$m = \pm 1$$

$$\Rightarrow m = 0, 0, 0, 1, -1$$

$$\therefore C.F. = (C_1 + C_2x + C_3x^2)e^{0.x} + C_4e^{1.x} + C_5e^{-1.x}$$

$$\therefore y = C.F. = C_1 + C_2x + C_3x^2 + C_4e^x + C_5e^{-x} \quad \underline{\text{Ans}}$$

(ix) $(D^2+4)(D^2+2D+1)y=0$

Solⁿ:

The A.E. is,

$$(m^2+4)(m^2+2m+1)=0$$

$$\Rightarrow (m^2+4)(m+1)^2=0$$

$$\Rightarrow (m^2+4)(m+1)(m+1)=0$$

$$\Rightarrow m^2=-4, m=-1, m=-1$$

$$\Rightarrow m^2=4i^2, m=-1, m=-1$$

$$\Rightarrow m=\pm 2i, m=-1, m=-1$$

$$\Rightarrow m=\pm 2i, -1, -1$$

$$\therefore \text{C.F.} = e^{0 \cdot x} [C_1 \cos 2x + C_2 \sin 2x] + (C_3 + C_4 x) e^{-1 \cdot x}$$

$$\therefore y = C_1 \cos 2x + C_2 \sin 2x + C_3 e^{-x} + C_4 x e^{-x} \quad \underline{\text{Ans}}$$

(x) $\frac{d^2 y}{dx^2} + y = 0$, given $y(0) = 2$, $y(\pi/2) = -2$

Solⁿ:

$$(D^2+1)y=0$$

The A.E. is,

$$m^2+1=0$$

$$\Rightarrow m^2=-1$$

$$\Rightarrow m=\pm i$$

$$\text{C.F.} = e^{0 \cdot x} [C_1 \cos x + C_2 \sin x]$$

$$\therefore y = C_1 \cos x + C_2 \sin x \quad \text{--- (1)}$$

when $x=0$ & $y=2$,
 $C_1 = 2$

when $x = \pi/2$ & $y = -2$,
 $C_2 = -2$

Putting in eqⁿ. (1), we get,

- $y = 2 \cos x - 2 \sin x = 2(\cos x - \sin x)$ Ans

Rules to Find Particular Integral (P.I.)

$$f(D)y = Q$$

$$P.I. = \frac{1}{f(D)} \cdot Q$$

Case-I: When $Q = e^{ax}$,

$$P.I. = \frac{1}{f(D)} \cdot e^{ax} = \frac{1}{f(a)} \cdot e^{ax}, \text{ provided } f(a) \neq 0$$

If $f(a) = 0$, it is case of failure.

$$P.I. = \frac{1}{f(D)} \cdot e^{ax} = x \frac{1}{f'(D)} \cdot e^{ax} = \frac{1}{f'(a)} \cdot e^{ax}, f'(a) \neq 0$$

Case - II: When $Q = \cos(ax+b)$ or $\sin(ax+b)$.

$$P.I. = \frac{1}{f(D^2)} \cdot \cos(ax+b) = \frac{1}{f(-a^2)} \cdot \cos(ax+b)$$

$$[f(-a^2) \neq 0]$$

$$P.I. = \frac{1}{f(D^2)} \cdot \sin(ax+b) = \frac{1}{f(-a^2)} \cdot \sin(ax+b)$$

$$[f(-a^2) \neq 0]$$

Case - III: When $Q = a^m$ or a polynomial of degree 'm' be positive integer.

① Take out the lowest degree term from $f(D)$ to reduce it to the form $[1 \pm F(D)]$

② Take it to the numerator which is expanded with the help of binomial expansions.

$$[1 \pm F(D)]^{-1}$$

$$(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

$$(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

Remark: If Q is x^m or polynomial of 'm', beyond the m^{th} power D need not be written because their higher derivative are zero.

Case-IV: When $Q = e^{ax} \cdot V$, where V is any function of x .

$$P.I. = \frac{1}{f(D)} \cdot e^{ax} V = e^{ax} \cdot \frac{1}{f(D+a)} \cdot V$$

Case-V: When $Q = xV$, where V is any function of x .

$$P.I. = \frac{1}{f(D)} \cdot xV = x \frac{1}{f(D)} \cdot V - \frac{f'(D)}{[f(D)]^2} \cdot V$$

Case-VI: When Q is any function of x .

$$P.I. = \frac{1}{D-a} \cdot Q = e^{ax} \int e^{-ax} Q \, dx$$

$$= \frac{1}{D+a} \cdot Q = e^{-ax} \int e^{ax} Q \, dx$$

Case-VII: When $Q = x^n \cdot \cos ax$ or $Q = x^n \cdot \sin ax$,
 $x^n \cdot e^{iax} = x^n [\cos ax + i \sin ax] = x^n \cos ax + i x^n \sin ax$

$$P.I. = \frac{1}{f(D)} \cdot x^n \cos ax = R.P. \cdot \frac{1}{f(D)} \cdot x^n e^{iax}$$

$$P.I. = \frac{1}{f(D)} \cdot x^n \sin ax = I.P. \cdot \frac{1}{f(D)} \cdot x^n \cdot e^{iax}$$

Q) Solve:

$$(i) \frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 5e^{3x}$$

Solⁿ:

$$[D^2 + 6D + 9]y = 5e^{3x}$$

The A.E. is,

$$m^2 + 6m + 9 = 0$$

$$\Rightarrow (m+3)^2 = 0$$

$$\Rightarrow (m+3)(m+3) = 0$$

$$\Rightarrow m = -3, -3$$

$$C.F. = (C_1 + C_2 x)e^{-3x}$$

$$P.I. = \frac{5e^{3x}}{D^2 + 6D + 9}$$

$$= 5 \times \frac{1}{3^2 + 6 \times 3 + 9} \times e^{3x}$$

$$= \frac{5}{36} \cdot e^{3x}$$

$$\therefore y = C.F. + P.I. = (C_1 + C_2 x)e^{-3x} + \frac{5}{36} \cdot e^{3x} \quad \underline{\underline{Ans}}$$

$$(ii) (D-2)^3 y = 17.e^{2x}$$

Solⁿ:

The A.E. is,

$$(m-2)^3 = 0$$

$$\Rightarrow (m-2)(m-2)(m-2) = 0$$

$$\Rightarrow m = 2, 2, 2$$

$$C.F. = (C_1 + C_2 x + C_3 x^2) e^{2x}$$

$$P.I. = \frac{1}{(D-2)^3} \cdot 17 e^{2x}$$

$$= 17 \cdot \frac{1}{(D-2)^3} \cdot e^{2x}, f(2) = 0$$

Case of failure,

$$P.I. = 17x \times \frac{1}{3(D-2)^2} \times e^{2x}, f(2) = 0$$

$$= 17x^2 \times \frac{1}{6(D-2)} \times e^{2x}, f(2) = 0$$

$$= 17x^3 \times \frac{1}{6} \times e^{2x}$$

$$= \frac{17}{6} x^3 e^{2x}$$

$$\therefore y = C.F. + P.I.$$

$$y = (C_1 + C_2 x + C_3 x^2) e^{2x} + \frac{17}{6} x^3 e^{2x} \quad \underline{\underline{\text{Ans}}}$$

$$(iii) (D^2 - 6D + 9)y = e^{2x} - \log 2$$

Solⁿ:

The A.E. is,

$$m^2 - 6m + 9 = 0$$

$$\Rightarrow (m-3)^2 = 0$$

$$\Rightarrow (m-3)(m-3) = 0$$

$$\Rightarrow m = 3, 3$$

$$C.F. = (C_1 + C_2 x) e^{3x}$$

$$P.I. = \frac{1}{(D^2 - 6D + 9)} \times (e^{2x} - \log 2)$$

$$= \frac{1}{(D^2 - 6D + 9)} \cdot e^{2x} - \frac{1}{(D^2 - 6D + 9)} \cdot \log 2$$

$$= \frac{1}{2^2 - 6 \times 2 + 9} \cdot e^{2x} - \log 2 \cdot \frac{1}{D^2 - 6D + 9} \cdot e^{0x}$$

$$= e^{2x} - \log 2 \cdot \frac{1}{0^2 + 6 \times 0 + 9} \cdot e^{0x}$$

$$= e^{2x} - \frac{1}{9} \cdot \log 2$$

Hence, the complete solution is,

$$y = (C_1 + C_2 x) e^{3x} + e^{2x} - \frac{1}{9} \log 2 \quad \underline{\text{Ans}}$$

(iv) $(D^3 - D^2 + 4D - 4)y = e^x$

Solⁿ:

The A.E. is,

$$m^3 - m^2 + 4m - 4 = 0$$

Using synthetic division,

$$\begin{array}{r|rrrr} 1 & 1 & -1 & 4 & -4 \\ & & 1 & 0 & 4 \\ \hline & 1 & 0 & 4 & 0 \end{array}$$

$$(m-1)(m^2 - 0 \cdot m + 4) = 0$$

$$\Rightarrow (m-1)(m^2 - 4) = 0$$

$$\Rightarrow m=1, m^2 = -4$$

$$\Rightarrow m=1, m = \pm 2i$$

$$C.F. = Ge^x + e^{0 \cdot x} [C_2 \cos 2x + C_3 \sin 2x]$$

$$P.I. = \frac{1}{(D^3 - D^2 + 4D - 4)} \cdot e^x$$

■ $f(1) = 0$, case of failure

$$P.I. = x \cdot \frac{1}{3D^2 - 2D + 4} \cdot e^x$$

$$= x \cdot \frac{1}{3(1)^2 - 2(1) + 4} \cdot e^x$$

$$\sinh ax = \frac{1}{a} (e^{ax} - e^{-ax})$$

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$$= \frac{1}{5} x e^x$$

$$\therefore y = C_1 e^x + C_2 \cos 2x + C_3 \sin 2x + \frac{1}{5} x e^x \quad \underline{\text{Ans}}$$

$$(v) (D^2 + 4D + 4)y = \sinh 2x$$

Solⁿ:

The A.E. is,

$$m^2 + 4m + 4 = 0$$

$$\Rightarrow (m+2)^2 = 0$$

$$\Rightarrow (m+2)(m+2) = 0$$

$$\Rightarrow m = -2, -2$$

$$C.F. = (C_1 + C_2 x) e^{-2x}$$

$$P.I. = \frac{1}{(D^2 + 4D + 4)} \cdot \sinh 2x$$

$$= \frac{1}{(D^2 + 4D + 4)} \cdot \frac{1}{2} (e^{2x} - e^{-2x})$$

$$= \frac{1}{2} \left[\frac{1}{(D^2 + 4D + 4)} e^{2x} - \frac{1}{(D^2 + 4D + 4)} e^{-2x} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2^2 + 4 \times 2 + 4} e^{2x} - \frac{1}{(-2)^2 + 4 \times (-2) + 4} e^{-2x} \right]$$

$$= \frac{1}{2} \left[\frac{1}{16} e^{2x} - \frac{1}{0} \cdot e^{-2x} \right]$$

Case of failure,

$$P.I. = \frac{1}{2} \left[\frac{1}{16} e^{2x} - x \cdot \frac{1}{2D+4} \cdot e^{-2x} \right]$$

Case of failure,

$$P.I. = \frac{1}{2} \left[\frac{1}{16} e^{2x} - x^2 \cdot \frac{1}{2} \cdot e^{-2x} \right]$$

$$= \frac{1}{32} e^{2x} - \frac{1}{4} x^2 \cdot e^{-2x}$$

$$\therefore y = C.F. + P.I.$$

$$= (C_1 + C_2 x) e^{-2x} + \frac{1}{32} e^{2x} - \frac{1}{4} x^2 \cdot e^{-2x} \quad \underline{\text{Ans}}$$

(vi) $(D^3 - 2D^2 - 5D + 6)y = e^{3x}$

Solⁿ:

The A.E. is,

$$m^3 - 2m^2 - 5m + 6 = 0$$

$$\Rightarrow (m-1)(m^2 - m - 6) = 0$$

$$\cancel{\Rightarrow (m-1)[m(m-1)]}$$

$$\Rightarrow m = 1, m = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times (-6)}}{2 \times 1}$$

1	1	-2	-5	6
		1	-1	-6
	1	-1	-6	0

$$\Rightarrow m = \frac{1 \pm 5}{2}, m = 1$$

$$\Rightarrow m = 1, 3, -2$$

$$C.F. = C_1 e^x + C_2 e^{3x} + C_3 e^{-2x}$$

$$P.I. = \frac{1}{(D^3 - 2D^2 - 5D + 6)} \cdot e^{3x}$$

$f(3) = 0$, case of failure

$$P.I. = x \cdot \frac{1}{(3D^2 - 4D - 5)} \cdot e^{3x}$$

$$= x \cdot \frac{1}{3 \times 3^2 - 4 \times 3 - 5} \cdot e^{3x}$$

$$= \frac{1}{10} \cdot x \cdot e^{3x}$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{3x} + C_3 e^{-2x} + \frac{1}{10} x e^{3x} \quad \underline{\text{Ans}}$$

$$(vii) (D^2 + a^2)y = \sin ax$$

Solⁿ:

The A.E. is,

$$m^2 + a^2 = 0$$

$$\Rightarrow m^2 = -a^2$$

$$\Rightarrow m^2 = a^2 \cdot i^2$$

$$\Rightarrow m = \pm ai$$

$$\begin{aligned} \text{C.F.} &= e^{0 \cdot x} [C_1 \cos ax + C_2 \sin ax] \\ &= C_1 \cos ax + C_2 \sin ax \end{aligned}$$

$$\text{P.I.} = \frac{1}{(D^2 + a^2)} \cdot \sin ax$$

$$= \frac{1}{-a^2 + a^2} \cdot \sin ax \quad [\because D^2 = -a^2]$$

Case of failure,

$$\text{P.I.} = x \cdot \left[\frac{1}{2D} \sin ax \right]$$

$$= \frac{x}{2} \int \sin ax \, dx$$

$$= -\frac{x}{2} \cdot \frac{\cos ax}{a}$$

$$\therefore y = C_1 \cos ax + C_2 \sin ax - \frac{x}{2a} \cos ax \quad \underline{\underline{\text{Ans}}}$$

(viii) $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 6y = \sin 3x$

Solⁿ:

$$[D^2 + 5D - 6]y = \sin 3x$$

The A.E. is,

$$m^2 + 5m - 6 = 0$$

$$\Rightarrow m^2 + (6-1)m - 6 = 0$$

$$\Rightarrow m^2 + 6m - m - 6 = 0$$

$$\Rightarrow m(m+6) - 1(m+6) = 0$$

$$\Rightarrow (m-1)(m+6) = 0$$

$$\Rightarrow m = 1, -6$$

$$C.F. = C_1 e^x + C_2 e^{-6x}$$

$$P.I. = \frac{1}{D^2 + 5D - 6} \cdot \sin 3x$$

$$= \frac{1}{-9 + 5D - 6} \cdot \sin 3x$$

$$= \frac{1}{5D - 15} \cdot \sin 3x$$

$$= \frac{1}{5(D-3)} \cdot \sin 3x$$

$$= \frac{1}{5} \frac{(D+3)}{(D-3)(D+3)} \sin 3x$$

$$= \frac{1}{5} \cdot \frac{(D+3)}{D^2-9} \sin 3x$$

$$= \frac{1}{5} \cdot \frac{(D+3)}{-9-9} \sin 3x$$

$$= \frac{-1}{90} [D \sin 3x + 3 \sin 3x]$$

$$~~= \frac{-1}{90} [3 \sin 3x]~~$$

$$= \frac{-1}{90} [3 \cos 3x + 3 \sin 3x]$$

$$= \frac{-1}{30} [\cos 3x + \sin 3x]$$

$$\therefore y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{-6x} - \frac{1}{30} (\cos 3x + \sin 3x)$$

(ix) $(D^2 + 5D - 6)y = \sin 3x + \cos 2x$

Solⁿ:

The A.E. is,

$$m^2 + 5m - 6 = 0$$

$$\Rightarrow m^2 + (6-1)m - 6 = 0$$

$$\Rightarrow m^2 + 6m - m - 6 = 0$$

$$\Rightarrow m(m+6) - 1(m+6) = 0$$

$$\Rightarrow (m-1)(m+6) = 0$$

$$\Rightarrow m = 1, -6$$

$$C.F. = C_1 e^x + C_2 e^{-6x}$$

$$P.I. = \frac{1}{(D^2 + 5D - 6)} \cdot (\sin 3x + \cos 2x)$$

$$= \frac{1}{(D^2 + 5D - 6)} \cdot \sin 3x + \frac{1}{(D^2 + 5D - 6)} \cdot \cos 2x$$

$$= \frac{1}{-(9) + 5D - 6} \cdot \sin 3x + \frac{1}{-4 + 5D - 6} \cdot \cos 2x$$

$$= \frac{1}{5D - 15} \cdot \sin 3x + \frac{1}{5D - 10} \cdot \cos 2x$$

$$= \frac{1}{5(D-3)} \cdot \sin 3x + \frac{1}{5(D-2)} \cdot \cos 2x$$

$$= \frac{1}{5} \cdot \frac{(D+3)}{(D-3)(D+3)} \cdot \sin 3x + \frac{1}{5} \cdot \frac{(D+2)}{(D-2)(D+2)} \cdot \cos 2x$$

$$= \frac{1}{5} \cdot \frac{(D+3)}{(D^2-9)} \cdot \sin 3x + \frac{1}{5} \cdot \frac{(D+2)}{(D^2-4)} \cdot \cos 2x$$

$$= \frac{1}{5} \cdot \frac{(D \sin 3x + 3 \sin 3x)}{(-9-9)} + \frac{1}{5} \cdot \frac{(D \cos 2x + 2 \cos 2x)}{-4-4}$$

$$= \frac{-1}{90} [3 \cos 3x + 3 \sin 3x] - \frac{1}{40} [-2 \sin 2x + 2 \cos 2x]$$

$$= \frac{-1}{30} (\cos 3x + \sin 3x) - \frac{1}{20} (\cos 2x - \sin 2x)$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{-6x} - \frac{1}{30} (\cos 3x + \sin 3x) - \frac{1}{20} (\cos 2x - \sin 2x). \quad \underline{\text{Ans}}$$

$$(x) [D^2 + 4]y = \sin 3x + \cos 2x$$

Solⁿ:

The A.E. is,

$$m^2 + 4 = 0$$

$$\Rightarrow m^2 = -4$$

$$\Rightarrow m^2 = 4i^2$$

$$\Rightarrow m = \pm 2i$$

$$\begin{aligned} \text{C.F.} &= e^{0x} [C_1 \cos 2x + C_2 \sin 2x] \\ &= C_1 \cos 2x + C_2 \sin 2x \end{aligned}$$

$$\text{P.I.} = \frac{1}{(D^2 + 4)} \cdot (\sin 3x + \cos 2x)$$

$$= \frac{1}{-9 + 4} \cdot \sin 3x + \frac{1}{-4 + 4} \cdot \cos 2x$$

$$= -\frac{1}{5} \sin 3x + x \frac{1}{20} \cos 2x = \frac{1}{20} [4x \cos 2x - 5 \sin 3x] \quad (*)$$

$$= -\frac{1}{5} \sin 3x + \frac{x}{2} \int \cos 2x \, dx$$

$$= -\frac{1}{5} \sin 3x + \frac{x}{2} \cdot \frac{\sin 2x}{2}$$

$$= -\frac{1}{5} \sin 3x + \frac{x}{4} \sin 2x$$

Hence, the complete solution is,

$$y = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{5} \sin 3x + \frac{x}{4} \sin 2x.$$

(xi) $(D^2 + 1)y = \sin x \cdot \sin 2x$

Soln:

The A.E. is,

$$m^2 + 1 = 0$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m^2 = i^2$$

$$\Rightarrow m = \pm i$$

$$\begin{aligned} \text{C.F.} &= e^{0x} [C_1 \cos x + C_2 \sin x] \\ &= C_1 \cos x + C_2 \sin x \end{aligned}$$

$$2\sin A \cdot \sin B = \cos(A-B) - \cos(A+B)$$

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$$P.I. = \frac{1}{(D^2+1)} \cdot \sin x \cdot \sin 2x$$

$$= \frac{1}{(D^2+1)} \cdot \frac{1}{2} \cdot (2 \sin 2x \sin x)$$

$$= \frac{1}{2} \cdot \frac{1}{(D^2+1)} \cdot \cancel{\sin 2x} [\cos(2x-x) - \cos(2x+x)]$$

$$= \frac{1}{2} \left[\frac{1}{(D^2+1)} \cdot \cos x - \frac{1}{(D^2+1)} \cdot \cos 3x \right]$$

$$= \frac{1}{2} \left[\frac{1}{-1+1} \cos x - \frac{1}{-9+1} \cos 3x \right]$$

$$= \frac{1}{2} \left[x \cdot \frac{1}{2D} \cos x + \frac{1}{8} \cos 3x \right]$$

$$= \frac{1}{2} \left[x \cdot \frac{1}{2} \sin x + \frac{1}{8} \cos 3x \right]$$

$$= \frac{1}{4} x \cdot \sin x + \frac{1}{16} \cos 3x$$

Hence, $y = C.F. + P.I.$

$$= C_1 \cos x + C_2 \sin x + \frac{1}{4} x \cdot \sin x + \frac{1}{16} \cos 3x.$$

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xii) $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$

Solⁿ:

The A.E. is,

$$m^3 - 3m^2 + 4m - 2 = 0$$

$$\Rightarrow (m-1)(m^2 - 2m + 2) = 0$$

$$\Rightarrow \cancel{(m-1)}[m^2 - 2m + 2]$$

$$\Rightarrow m = 1, m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times 2}}{2 \times 1}$$

$$= \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{2 \pm \sqrt{4i^2}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$\Rightarrow m = 1, 1 \pm i$$

$$C.F. = C_1 e^x + e^x [C_2 \cos x + C_3 \sin x]$$

$$P.I. = \frac{1}{(D^3 - 3D^2 + 4D - 2)} e^x + \frac{1}{(D^3 - 3D^2 + 4D - 2)} \cos x$$

$$= x \cdot \frac{1}{(3D^2 - 6D + 4)} e^x + \frac{1}{[D(-1) - 3(-1) + 4D - 2]} \cos x$$

$$= x \cdot e^x + \frac{1}{(3D + 1)} \cos x$$

$$= x e^x + \frac{(3D - 1)}{(3D + 1)(3D - 1)} \cos x$$

$$= x e^x + \frac{(3D-1)}{9D^2-1} \cos x$$

$$= x e^x + \frac{(3D-1)}{9(-1)-1} \cos x$$

$$= x e^x - \frac{1}{10} [3D \cos x - \cos x]$$

$$= x e^x - \frac{1}{10} [-3 \sin x - \cos x]$$

$$= x e^x + \frac{1}{10} (3 \sin x + \cos x)$$

Hence, $y = C.F. + P.I.$

$$= C_1 e^x + e^x [C_2 \cos x + C_3 \sin x] + x e^x + \frac{1}{10} (3 \sin x + \cos x).$$

(xiii) $[D^2 + 5D - 6]y = \sin 4x \cdot \sin x$

Solⁿ:

The A.E. is,

$$m^2 + 5m - 6 = 0$$

$$\Rightarrow m^2 + 6m - m - 6 = 0$$

$$\Rightarrow m(m+6) - 1(m+6) = 0$$

$$\Rightarrow (m-1)(m+6) = 0$$

$$\Rightarrow m = 1, -6$$

$$C.F. = C_1 e^x + C_2 e^{-6x}$$

$$P.I. = \frac{1}{(D^2 + 5D - 6)} \cdot \sin 4x \cdot \sin x$$

$$= \frac{1}{(D^2 + 5D - 6)} \cdot \frac{1}{2} [2 \sin 4x \cdot \sin x]$$

$$= \frac{1}{2} \cdot \frac{1}{(D^2 + 5D - 6)} [\cos(4x - x) - \cos(4x + x)]$$

$$= \frac{1}{2} \left[\frac{1}{(D^2 + 5D - 6)} \cdot \cos 3x - \frac{1}{(D^2 + 5D - 6)} \cdot \cos 5x \right]$$

$$= \frac{1}{2} \left[\frac{1}{(-9 + 5D - 6)} \cdot \cos 3x - \frac{1}{(-25 + 5D - 6)} \cdot \cos 5x \right]$$

$$= \frac{1}{2} \left[\frac{1}{(5D - 15)} \cdot \cos 3x - \frac{1}{(5D - 31)} \cdot \cos 5x \right]$$

$$= \frac{1}{2} \left[\frac{1}{5(D - 3)} \cdot \cos 3x - \frac{1}{(5D - 31)} \cdot \cos 5x \right]$$

$$= \frac{1}{2} \left[\frac{(D + 3)}{5(D - 3)(D + 3)} \cdot \cos 3x - \frac{(5D + 31)}{(5D - 31)(5D + 31)} \cdot \cos 5x \right]$$

$$= \frac{1}{2} \left[\frac{(D + 3) \cos 3x}{5(D^2 - 9)} - \frac{(5D + 31) \cos 5x}{(25D^2 - 961)} \right]$$

$$= \frac{1}{2} \left[\frac{D(\cos 3x) + 3 \cos 3x}{5(-9-9)} - \frac{5D(\cos 5x) + 31 \cos 5x}{25 \times (-25) - 961} \right]$$

$$= \frac{1}{2} \left[\frac{-1}{90} [-3 \sin 3x + 3 \cos 3x] + \frac{1}{1586} [5 \times 5 \sin 5x + 31 \cos 5x] \right]$$

$$= \cancel{\frac{-1}{180} [-3 \cos 3x - \sin 3x]} + \cancel{\frac{1}{3172} [15 \cos 5x - 25 \sin 5x]}$$

$$= \frac{1}{60} [\sin 3x - \cos 3x] - \frac{1}{3172} [25 \sin 5x - 31 \cos 5x]$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{-6x} + \frac{1}{60} [\sin 3x - \cos 3x] - \frac{1}{3172} [25 \sin 5x - 31 \cos 5x]$$

Ans

$$(xiv) \frac{d^2 y}{dx^2} + y = x^2$$

Solⁿ:

$$[D^2 + 1]y = x^2$$

The A.E. is,

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i$$

$$C.F. = C_1 \cos x + C_2 \sin x$$

$$P.I. = \frac{1}{(D^2 + 1)} \cdot x^2$$

$$= \frac{1}{1 + D^2} x^2$$

$$= [1 + D^2]^{-1} x^2$$

$$= [1 - D^2 + D^4 \dots] x^2$$

Leaving higher power of D ,

$$P.I. = [1 - D^2] x^2$$

$$= x^2 - D^2 x^2$$

$$= x^2 - 2$$

Hence, $y = C.F. + P.I.$

$$= C_1 \cos x + C_2 \sin x + x^2 - 2.$$

(xv) $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = x$

Solⁿ:

$$[D^2 + 2D + 1]y = x$$

The A.E. is,

$$m^2 + 2m + 1 = 0$$

$$\Rightarrow (m+1)^2 = 0$$

$$\Rightarrow m = -1, -1$$

$$C.F. = (C_1 + C_2 x)e^{-x}$$

$$P.I. = \frac{1}{(D^2 + 2D + 1)} \cdot x$$

$$= \frac{1}{(D+1)^2} \cdot x$$

$$= [1+D]^{-2} x = [1 - 2D + 3D^2 - \dots]x$$

Leaving higher order of D ,

$$P.I. = [1 - 2D]x$$

$$= x - 2Dx$$

$$= x - 2$$

Hence, $y = C.F. + P.I.$

$$= (C_1 + C_2 x)e^{-x} + x - 2$$

(xvi) $\frac{d^2 y}{dx^2} + y = x^2 + 2x + 1$

Solⁿ:

$$[D^2 + 1]y = x^2 + 2x + 1$$

The A.E. is,

$$m^2 + 1 = 0$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m = \pm i$$

$$C.F. = e^{0x} [C_1 \cos x + C_2 \sin x]$$

$$P.I. = \frac{1}{(D^2 + 1)} \cdot (x^2 + 2x + 1)$$

$$= [1 + D^2]^{-1} (x^2 + 2x + 1)$$

$$= [1 - D^2 + D^4 - \dots] (x^2 + 2x + 1)$$

$$= [1 - D^2] (x^2 + 2x + 1)$$

$$= x^2 + 2x + 1 - 2$$

$$= x^2 + 2x - 1$$

Hence, $y = C.F. + P.I.$

$$= C_1 \cos x + C_2 \sin x + x^2 + 2x - 1.$$

(xvii) $[D^2 + D - 2]y = x$

Solⁿ:

The A.E. is,

$$m^2 + m - 2 = 0$$

$$\Rightarrow m^2 + (2-1)m - 2 = 0$$

$$\Rightarrow m^2 + 2m - m - 2 = 0$$

$$\Rightarrow m(m+2) - 1(m+2) = 0$$

$$\Rightarrow (m-1)(m+2) = 0$$

$$\Rightarrow m = 1, -2$$

$$C.F. = C_1 e^x + C_2 e^{-2x}$$

$$P.I. = \frac{1}{(D^2 + D - 2)} \cdot x$$

$$= \frac{1}{(-2 + D - 2)} \cdot x$$

$$= \frac{1}{(-2 + D + D^2)} \cdot x$$

$$= \frac{1}{-2 \left[1 - \frac{D}{2} - \frac{D^2}{2} \right]} \cdot x$$

$$= \frac{1}{-2 \left[1 - \left[\frac{D}{2} + \frac{D^2}{2} \right] \right]} x$$

$$= -\frac{1}{2} \left[1 - \left(\frac{D}{2} + \frac{D^2}{2} \right) \right]^{-1} x$$

$$= -\frac{1}{2} \left[1 + \left(\frac{D}{2} + \frac{D^2}{2} \right) + \left(\frac{D}{2} + \frac{D^2}{2} \right)^2 + \dots \right] x$$

leaving higher order of D ,

$$P.I. = -\frac{1}{2} \left[1 + \frac{D}{2} \right] x$$

$$= -\frac{1}{2} \left[x + \frac{1}{2} D x \right]$$

$$= -\frac{1}{2} \left[x + \frac{1}{2} \right]$$

$$= -\frac{1}{4} (2x+1)$$

$$\therefore y = C_1 e^x + C_2 e^{-2x} - \frac{1}{4} (2x+1)$$

$$(xviii) \frac{d^2 y}{dx^2} - y = x^2$$

Solⁿ:

$$[D^2 - 1] y = x^2$$

The A.E. is,

$$m^2 - 1 = 0$$

$$\Rightarrow m^2 = 1$$

$$\Rightarrow m = \pm 1$$

$$\Rightarrow m = 1, -1$$

$$C.F. = C_1 e^x + C_2 e^{-x}$$

$$P.I. = \frac{1}{(D^2 - 1)} \cdot x^2$$

$$= \frac{1}{-(1 - D^2)} \cdot x^2$$

$$= -[1 - D^2]^{-1} x^2$$

$$= -[1 + D^2 + D^4 + D^6 + \dots] x^2$$

• Leaving higher order of D ,

$$P.I. = -[1 + D^2] x^2$$

$$= -[x^2 + D^2(x^2)]$$

$$= -(x^2 + 2)$$

$$= -2 - x^2$$

$$= -x^2 - 2$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{-x} - x^2 - 2 \quad \underline{\text{Ans}}$$

(xix) $y'' - 2y' + 2y = e^x \cos x$

Solⁿ:

$$[D^2 - 2D + 2] y = e^x \cos x$$

The A.E. is,

$$m^2 - 2m + 2 = 0$$

$$\Rightarrow m = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i$$

$$C.F. = e^x [C_1 \cos x + C_2 \sin x]$$

$$P.I. = \frac{1}{(D^2 - 2D + 2)} \cdot e^x \cos x$$

$$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 2} \cdot \cos x$$

$$= e^x \frac{1}{D^2 + 1 + 2D - 2D - 2 + 2} \cos x$$

$$= e^x \frac{1}{D^2 + 1} \cdot \cos x$$

$$= e^x \cdot x \cdot \frac{1}{2D} \cdot \cos x$$

$$= \frac{1}{2} e^x \cdot x \cdot \sin x$$

Hence, the C.S. is,

$$y = e^x [C_1 \cos x + C_2 \sin x] + \frac{x e^x \sin x}{2}$$

$$(xx) [D^2 - 2D]y = e^x \cdot \sin x$$

Solⁿ:

The A.E. is,

$$m^2 - 2m = 0$$

$$\Rightarrow m(m-2) = 0$$

$$\Rightarrow m = 0, 2$$

$$C.F. = C_1 e^{0x} + C_2 e^{2x} = C_1 + C_2 e^{2x}$$

$$P.I = \frac{1}{(D^2 - 2D)} \cdot e^x \cdot \sin x$$

$$= \frac{1}{D(D-2)} \cdot e^x \cdot \sin x$$

$$= e^x \frac{1}{(D+1)(D+1-2)} \cdot \sin x$$

$$= e^x \frac{1}{(D^2 - 1)} \sin x$$

$$= e^x \cdot x \cdot \frac{1}{2D} \sin x$$

$$= \frac{x e^x}{2} (\cos x)$$

$$\therefore y = C_1 + C_2 e^{2x} - \frac{x e^x}{2} \cos x$$

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(xxi)

$$[D^2 - 2D + 1]y = x e^x \cdot \cos x$$

Solⁿ:

The A.E. is,

$$m^2 - 2m + 1 = 0$$

$$\Rightarrow (m-1)^2 = 0$$

$$\Rightarrow m = 1, 1$$

$$C.F. = (C_1 + C_2 x) e^x$$

$$P.I. = \frac{1}{(D^2 - 2D + 1)} \cdot x e^x \cdot \cos x$$

$$= \frac{1}{(D-1)^2} e^x \cdot x \cdot \cos x$$

$$= e^x \frac{1}{(D+1-1)^2} x \cdot \cos x$$

$$= e^x \cdot \frac{1}{D^2} x \cdot \cos x$$

$$= e^x \frac{1}{D} \left[\int \frac{x}{u} \frac{\cos x}{v} dx \right]$$

$$= e^x \frac{1}{D} \left[x(\sin x) - (1)(-\cos x) \right] \quad [\because \int uv = uv_1 - u'v_2 \dots]$$

$$= e^x \frac{1}{D} [x \sin x + \cos x]$$

$$= e^x \left[\int x \sin x \, dx + \int \cos x \, dx \right]$$

$$= e^x [x(-\cos x) - 1(-\sin x) + \sin x]$$

$$= e^x [-x \cos x + \sin x + \sin x]$$

$$= e^x [2 \sin x - x \cos x]$$

$$\therefore y = C.F. + P.I.$$

$$= (C_1 + C_2 x) e^x + e^x [2 \sin x - x \cos x]$$

$$(xxii) [D^2 - 2D + 1]y = x e^x \sin x$$

Solⁿ:

The A.E. is,

$$m^2 - 2m + 1 = 0$$

$$\Rightarrow m^2 - m - m + 1 = 0$$

$$\Rightarrow m(m-1) - 1(m-1) = 0$$

$$\Rightarrow (m-1)(m-1) = 0$$

$$\Rightarrow m = 1, 1$$

$$C.F. = (C_1 + C_2 x) e^x$$

$$P.I. = \frac{1}{(D^2 - 2D + 1)} x e^x \sin x$$

$$= \frac{1}{(D-1)^2} x e^x \sin x$$

$$= e^x \cdot \frac{1}{(D+1-1)^2} x \sin x$$

$$= e^x \frac{1}{D^2} x \sin x$$

$$= e^x \frac{1}{D} \left[\int x \sin x \, dx \right]$$

$$= e^x \frac{1}{D} [x(-\cos x) - (1)(-\sin x)]$$

$$= e^x \frac{1}{D} (\sin x - x \cos x)$$

$$= e^x \left[\int \sin x \, dx - \int x \cos x \, dx \right]$$

$$= e^x \left[-\cos x - [x(\sin x) - (1)(-\cos x)] \right]$$

$$= e^x [-\cos x - x \sin x + \cos x]$$

$$= e^x [-x \sin x - 2 \cos x]$$

Hence, $y = C.F. + P.I.$

$$= (C_1 + C_2 x)e^x - x e^x \sin x - 2e^x \cos x.$$

(xxiii) $[D^2 - 3D + 2] y = x e^{3x}$

Solⁿ:

The A.E. is,

$$m^2 - 3m + 2 = 0$$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m-2) - 1(m-2) = 0$$

$$\Rightarrow (m-1)(m-2) = 0$$

$$\Rightarrow m = 1, 2$$

$$C.F. = C_1 e^x + C_2 e^{2x}$$

$$P.I. = \frac{1}{(D^2 - 3D + 2)} \cdot x e^{3x}$$

$$= e^{3x} \frac{1}{(D+3)^2 - 3(D+3) + 2} \cdot x$$

$$= e^{3x} \frac{1}{D^2 + 9 + 6D - 3D - 9 + 2} \cdot x$$

$$= e^{3x} \frac{1}{D^2 + 3D + 2} \cdot x$$

$$= e^{3x} \frac{1}{2 + 3D + D^2} \cdot x$$

$$= e^{3x} \frac{1}{2 \left[1 + \left(\frac{3D}{2} + \frac{D^2}{2} \right) \right]} \cdot x$$

$$= \frac{e^{3x}}{2} \left[1 + \left(\frac{3D}{2} + \frac{D^2}{2} \right) \right]^{-1} x$$

$$= \frac{e^{3x}}{2} \left[1 - \left(\frac{3D}{2} + \frac{D^2}{2} \right) + \dots \right] x$$

Leaving higher power of D

$$P.I. = \frac{e^{3x}}{2} \left[1 - \frac{3D}{2} \right] x$$

$$= \frac{e^{3x}}{2} \left[x - \frac{3}{2} D(x) \right]$$

$$= \frac{e^{3x}}{2} \left[x - \frac{3}{2} \right]$$

$$= \frac{e^{3x}}{4} (2x - 3)$$

Hence, $y = C.F. + P.I.$

$$= C_1 e^x + C_2 e^{2x} + \frac{e^{3x}}{4} (2x - 3)$$

$$(xxiv) [D^2 - 4D + 3]y = 2x e^{3x}$$

Solⁿ:

The A.E. is,

$$m^2 - 4m + 3 = 0$$

$$\Rightarrow m^2 - 3m - m + 3 = 0$$

$$\Rightarrow m(m-3) - 1(m-3) = 0$$

$$\Rightarrow (m-1)(m-3) = 0$$

$$\Rightarrow m = 1, 3$$

$$C.F. = C_1 e^x + C_2 e^{3x}$$

$$P.I. = \frac{1}{(D^2 - 4D + 3)} \cdot 2x e^{3x}$$

$$= e^{3x} \cdot \frac{1}{(D+3)^2 - 4(D+3) + 3} \cdot 2x$$

$$= e^{3x} \cdot \frac{1}{(D^2 + 9 + 6D - 4D - 12 + 3)} \cdot 2x$$

$$= e^{3x} \cdot \frac{1}{(D^2 + 2D)} \cdot 2x$$

$$= e^{3x} \cdot \frac{1}{2D(1+D)} \cdot 2x$$

$$= e^{3x} \cdot \frac{1}{2D} [1+D]^{-1} \cdot 2x$$

$$= e^{3x} \cdot \frac{1}{2D} \left[1 - \frac{D}{2} + \frac{D^2}{2} - \frac{D^3}{2} + \dots \right] 2x$$

Leaving higher power of D ,

$$P.I. = e^{3x} \cdot \frac{1}{2D} \left[1 - \frac{D}{2} \right] 2x$$

$$= e^{3x} \cdot \frac{1}{2D} \left[2x - \frac{2D(x)}{2} \right]$$

$$= e^{3x} \cdot \frac{1}{2} \int (2x - 1) dx$$

$$= \frac{e^{3x}}{2} \cdot \frac{x^2}{2} - \frac{e^{3x}}{2} x$$

Hence, the C.S is,

$$y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{3x} + \frac{e^{3x}}{2} (x^2 - x).$$

(xxv) $[D^2 + 2D + 1]y = x \cos x$

Solⁿ:

The A.E. is,

$$m^2 + 2m + 1 = 0$$

$$\Rightarrow (m+1)^2 = 0$$

$$\Rightarrow m = -1, -1$$

$$C.F. = (C_1 + C_2 x) e^{-x}$$

$$P.I. = \frac{1}{(D^2 + 2D + 1)} \cdot x \cos x$$

$$= x \frac{1}{(D^2 + 2D + 1)} \cdot \cos x - \frac{(2D + 2)}{(D^2 + 2D + 1)^2} \cdot \cos x$$

$$= x \cdot \frac{1}{-1 + 2D + 1} \cdot \cos x - \frac{2(D + 1)}{(-1 + 2D + 1)^2} \cos x$$

$$= x \cdot \frac{1}{2D} \cos x - \frac{2(D + 1)}{4D^2} \cos x$$

$$= \frac{x}{2} \int \cos x \, dx - \frac{2(D + 1)}{4x - 1} \cos x$$

$$= \frac{x}{2} \cdot \sin x + \frac{1}{2} [D(\cos x) + \cos x]$$

$$= \frac{x}{2} \sin x + \frac{1}{2} [-\sin x + \cos x]$$

$$= \frac{1}{2} (x-1) \sin x + \frac{1}{2} \cos x$$

Hence, the complete solution is,

$$y = (C_1 + C_2 x) e^{-x} + \frac{1}{2} (x-1) \sin x + \frac{1}{2} \cos x \quad \underline{\underline{\text{Ans}}}$$

$$(xxvi) \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = x \sin x$$

Solⁿ:

$$(D^2 - 2D + 1)y = x \sin x$$

The A.E. is,

$$m^2 - 2m + 1 = 0$$

$$\Rightarrow (m-1)^2 = 0$$

$$\Rightarrow m = 1, 1$$

$$C.F. = (C_1 + C_2 x) e^x$$

$$P.I. = \frac{1}{(D^2 - 2D + 1)} \cdot x \sin x$$

$$= x \frac{1}{(D^2 - 2D + 1)} \cdot \sin x - \frac{(2D - 2)}{(D^2 - 2D + 1)^2} \sin x$$

$$= x \frac{1}{(-1 - 2D + 1)} \cdot \sin x - \frac{(2D - 2)}{(-1 - 2D + 1)^2} \sin x$$

$$= -\frac{x}{2} \int \sin x \, dx - \frac{2(D-1)}{4D^2} \sin x$$

$$= +\frac{x}{2} \cos x - \frac{1}{2x-1} [D(\sin x) - \sin x]$$

$$= \frac{x}{2} \cos x + \frac{1}{2} [\cos x - \sin x]$$

$$= \frac{1}{2} (x+1) \cos x - \frac{1}{2} \sin x$$

Hence, the C.S. is,

$$y = C.F. + P.I.$$

$$= (C_1 + C_2 x) e^x + \frac{1}{2} (x+1) \cos x - \frac{1}{2} \sin x \quad \underline{\underline{Ans}}$$

$$(xxvii) (D^2+4)y = 3x \sin x$$

Solⁿ:

The A.E. is,

$$m^2+4=0$$

$$\Rightarrow m^2 = -4$$

$$\Rightarrow m^2 = 4i^2$$

$$\Rightarrow m = \pm 2i$$

$$C.F. = e^{0x} [C_1 \cos 2x + C_2 \sin 2x]$$

$$P.I. = \frac{1}{(D^2+4)} \cdot 3x \sin x$$

$$= 3 \left[x \frac{1}{(D^2+4)} \sin x - \frac{2D}{(D^2+4)^2} \sin x \right]$$

$$= 3 \left[x \frac{1}{(-1+4)} \sin x - \frac{2D}{(-1+4)^2} \sin x \right]$$

$$= 3 \left[x \cdot \frac{1}{3} \cdot \sin x - \frac{2D}{9} \sin x \right]$$

$$= x \sin x - \frac{2}{3} D(\sin x)$$

$$= x \sin x - \frac{2}{3} \cos x$$

$$\therefore y = C.F. + P.I$$

$$= C_1 \cos 2x + C_2 \sin 2x + x \sin x - \frac{2}{3} \cos x$$

(xxviii) $\left(\frac{d^2 y}{dx^2} + 9y = \sec 3x \right)$

Solⁿ:

$$[D^2 + 9]y = \sec 3x$$

The A.E. is,

$$m^2 + 9 = 0$$

$$\Rightarrow m^2 = -9$$

$$\Rightarrow m = \pm 3i$$

$$C.F. = C_1 \cos 3x + C_2 \sin 3x$$

$$P.I. = \frac{1}{(D^2 + 9)} \cdot \sec 3x$$

$$= \frac{1}{(D^2 - 9i^2)} \cdot \sec 3x$$

$$= \frac{1}{(D - 3i)(D + 3i)} \cdot \sec 3x$$

$$= \frac{1}{6i} \left[\frac{1}{D - 3i} - \frac{1}{D + 3i} \right] \sec 3x$$

$$= \frac{1}{6i} \left[\frac{1}{(D - 3i)} \cdot \sec 3x - \frac{1}{(D + 3i)} \cdot \sec 3x \right]$$

Taking,

Date _____

$$\frac{1}{(D-3i)} \sec 3x = e^{3ix} \int e^{-3ix} \sec 3x \, dx$$

$$= e^{3ix} \int \left(\frac{\cos 3x - i \sin 3x}{\cos 3x} \right) dx$$

$$[\because e^{-iax} = \cos ax - i \sin ax]$$

$$= e^{3ix} \int (1 - i \tan 3x) \, dx$$

$$= e^{3ix} \left[x + \frac{i}{3} \log(\cos 3x) \right] \text{ --- (1)}$$

Replacing i by $-i$ in eqⁿ. (1),

$$\frac{1}{(D+3i)} \sec 3x = e^{-3ix} \left[x - \frac{i}{3} \log(\cos 3x) \right] \text{ --- (2)}$$

$$P.I. = \frac{1}{6i} \left[e^{3ix} \int x + \frac{i}{3} \log(\cos 3x) \right]$$

$$- e^{-3ix} \int x - \frac{i}{3} \log(\cos 3x) \right]$$

$$= \frac{1}{6i} \left[x(e^{3ix} - e^{-3ix}) + \frac{i}{3} \log(\cos 3x) [e^{3ix} + e^{-3ix}] \right]$$

$$\left[\frac{e^{iax} - e^{-iax}}{2i} = \sin ax, \quad \frac{e^{iax} + e^{-iax}}{2} = \cos ax \right]$$

$$P.I. = \frac{1}{3} \left[x \left(\frac{e^{3ix} - e^{-3ix}}{2i} \right) + i \log \cos 3x \left(\frac{e^{3ix} + e^{-3ix}}{2i} \right) \right]$$

$$= \frac{1}{3} \left[x \sin 3x + \frac{\cos 3x}{3} (\log \cos 3x) \right]$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 \cos 3x + C_2 \sin 3x + \frac{x}{3} \sin 3x$$

$$+ \frac{\cos 3x}{9} (\log \cos 3x) \quad \underline{\text{Ans}}$$

$$(xxix) (D^2 + a^2)y = \sec ax$$

Solⁿ:

The A.E. is,

$$m^2 + a^2 = 0$$

$$\Rightarrow m^2 = -a^2$$

$$\Rightarrow m = \pm ai$$

$$C.F. = [C_1 \cos ax + C_2 \sin ax]$$

$$P.I. = \frac{1}{(D^2 + a^2)} \sec ax$$

$$= \frac{1}{(D^2 - i^2 a^2)} \sec ax$$

$$= \frac{1}{(D + ai)(D - ai)} \sec ax$$

$$= \frac{1}{(D - ai)(D + ai)} \sec ax$$

$$= \frac{1}{2ai} \left[\frac{1}{(D - ai)} - \frac{1}{(D + ai)} \right] \sec ax$$

$$= \frac{1}{2ai} \left[\frac{1}{(D - ai)} \sec ax - \frac{1}{(D + ai)} \sec ax \right]$$

Taking,

$$\frac{1}{(D - ai)} \sec ax$$

$$= e^{iax} \int e^{-iax} \sec ax \, dx$$

$$= e^{iax} \int \left(\frac{\cos ax - i \sin ax}{\cos ax} \right) dx$$

$$= e^{iax} \int (1 - i \tan ax) \, dx$$

$$= e^{iax} \left[x + \frac{i}{a} \log \cos ax \right] \quad \text{--- (1)}$$

Replacing i with $-i$ in eqⁿ. (1),

$$\frac{1}{(D+ai)} \cdot \sec ax = e^{-iax} \left[x - \frac{i}{a} \log \cos ax \right] \quad \text{--- (2)}$$

$$P.I. = \frac{1}{2ai} \left[e^{iax} \left(x + \frac{i}{a} \log \cos ax \right) - \right.$$

$$\left. e^{-iax} \left(x - \frac{i}{a} \log \cos ax \right) \right]$$

$$= \frac{1}{2ai} \left[x(e^{iax} - e^{-iax}) + \frac{i}{a} \log \cos ax (e^{iax} + e^{-iax}) \right]$$

$$= \frac{1}{a} \left[x \left(\frac{e^{iax} - e^{-iax}}{2i} \right) + \frac{i}{a} \log \cos ax \left(\frac{e^{iax} + e^{-iax}}{2i} \right) \right]$$

$$= \frac{1}{a} \left[x \sin ax + \frac{1}{a} \log \cos ax (\cos ax) \right]$$

$$= \frac{x}{a} \sin ax + \frac{\cos ax}{a^2} (\log \cos ax)$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 \cos ax + C_2 \sin ax + \frac{x}{a} \sin ax + \frac{\cos ax}{a^2} (\log \cos ax).$$

$$(xxx) [D^2 + 1]y = \operatorname{cosec} x$$

Solⁿ:

The A.E. is is,

$$m^2 + 1 = 0$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m^2 = i^2$$

$$\Rightarrow m = \pm i$$

$$C.F. = C_1 \cos x + C_2 \sin x$$

$$P.I. = \frac{1}{(D^2 + 1)} \operatorname{cosec} x$$

$$= \frac{1}{(D^2 - i^2)} \operatorname{cosec} x$$

$$= \frac{1}{(D - i)(D + i)} \operatorname{cosec} x$$

$$= \frac{1}{2i} \left[\frac{1}{D-i} - \frac{1}{D+i} \right] \operatorname{cosec} x$$

$$= \frac{1}{2i} \left[\frac{1}{D-i} \operatorname{cosec} x - \frac{1}{D+i} \operatorname{cosec} x \right]$$

Taking,

$$\frac{1}{D-i} \operatorname{cosec} x$$

$$= e^{ix} \int e^{-ix} \operatorname{cosec} x \, dx$$

$$= e^{ix} \int \left(\frac{\cos x - i \sin x}{\sin x} \right) dx$$

$$= e^{ix} \int (\cot x - i) dx$$

$$= e^{ix} [\log \sin x - ix] \quad \text{--- (1)}$$

Replacing i with $-i$ in eqⁿ. (1),

$$\frac{1}{D+i} \operatorname{cosec} x = e^{-ix} [\log \sin x + ix] \quad \text{--- (2)}$$

$$P.I. = \frac{1}{2i} \left[e^{ix} [\log \sin x - ix] - e^{-ix} [\log \sin x + ix] \right]$$

$$= \frac{1}{2i} \left[\log \sin x (e^{ix} - e^{-ix}) - ix(e^{ix} + e^{-ix}) \right]$$

$$= \log \sin x \left(\frac{e^{ix} - e^{-ix}}{2i} \right) - ix \left(\frac{e^{ix} + e^{-ix}}{2i} \right)$$

$$= \sin x (\log \sin x) - x \cos x$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 \cos x + C_2 \sin x + \sin x (\log \sin x) - x \cos x.$$

Imp.

31. $[D^2 + 3D + 2]y = e^{e^x}$

Solⁿ:

The A.E. is,

$$m^2 + 3m + 2 = 0$$

$$\Rightarrow m^2 + 2m + m + 2 = 0$$

$$\Rightarrow m(m+2) + 1(m+2) = 0$$

$$\Rightarrow (m+1)(m+2) = 0$$

$$\Rightarrow m = -1, -2$$

$$C.F. = C_1 e^{-x} + C_2 e^{-2x}$$

$$P.I. = \frac{1}{(D^2 + 3D + 2)} \cdot e^{e^x}$$

$$= \frac{1}{(D+1)(D+2)} \cdot e^{ex}$$

$$= \left[\frac{A}{D+1} + \frac{B}{D+2} \right] e^{ex}$$

$$= \left[\frac{1}{D+1} - \frac{1}{D+2} \right] e^{ex}$$

$$= \frac{1}{D+1} \cdot e^{ex} - \frac{1}{D+2} \cdot e^{ex}$$

$$= P_1 - P_2$$

~~P_1 =~~
$$P_1 = \frac{1}{D+1} \cdot e^{ex} = e^{-x} \int e^x \cdot e^{ex} dx$$

$$\blacksquare \text{ Put } e^x = z \Rightarrow e^x dx = dz$$

$$P_1 = e^{-x} \int e^{2z} dz = e^{-x} e^z = e^{-x} e^{ex}$$

Also,

$$P_2 = \frac{1}{D+2} \cdot e^{ex} = e^{-2x} \int e^{2x} \cdot e^{ex} dx$$

$$= e^{-2x} \int e^x \cdot e^x \cdot e^{ex} dx$$

$$\text{Put } e^x = z \Rightarrow e^x dx = dz$$

$$P_2 = e^{-2x} \int \frac{z \cdot e^z}{\frac{u}{z}} dz$$

$$P_2 = e^{-2x} [ze^z - (1)(e^z)] \quad [\because \int u v dx = uv_1 - u'v_2 + \dots]$$

$$= e^{-2x} [e^x \cdot e^{e^x} - e^{e^x}]$$

$$= e^{-x} \cdot e^{e^x} - e^{-2x} \cdot e^{e^x}$$

$$\therefore P.I. = e^{-x} \cdot e^{e^x} - e^{-x} \cdot e^{e^x} + e^{-2x} \cdot e^{e^x} = e^{-2x} \cdot e^{e^x}$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 e^{-x} + C_2 e^{-2x} + e^{-2x} \cdot e^{e^x}$$

$$32. \frac{d^2 y}{dx^2} + a^2 y = \tan ax$$

Soln:

$$[D^2 + a^2] y = \tan ax$$

The A.E. is,

$$m^2 + a^2 = 0$$

$$\Rightarrow m^2 = -a^2$$

$$\Rightarrow m = \pm ia$$

$$C.F. = C_1 \cos ax + C_2 \sin ax$$

$$P.I. = \frac{1}{(D^2 + a^2)} \cdot \tan ax$$

$$= \frac{1}{D^2 - a^2 i^2} \cdot \tan ax$$

$$= \frac{1}{(D+ia)(D-ia)} \cdot \tan ax$$

$$= \left[\frac{A}{D-ia} + \frac{B}{D+ia} \right] \tan ax$$

$$= \frac{1}{2ia} \left[\frac{1}{D-ia} - \frac{1}{D+ia} \right] \tan ax$$

$$= \frac{1}{2ia} \left[\frac{1}{D-ia} \cdot \tan ax - \frac{1}{D+ia} \cdot \tan ax \right]$$

First part,

$$\frac{1}{D-ia} \cdot \tan ax$$

$$= e^{iax} \int e^{-iax} \tan ax \, dx$$

$$= e^{iax} \int (\cos ax - i \sin ax) \frac{\sin ax}{\cos ax} \, dx$$

$[\because e^{-iax} = \cos ax - i \sin ax]$

$$= e^{iax} \int \left[\sin ax - i \frac{\sin^2 ax}{\cos ax} \right] \, dx$$

$$= e^{iax} \int \left[\sin ax - i \frac{(1 - \cos^2 ax)}{\cos ax} \right] \, dx$$

$$= e^{iax} \int [\sin ax - i \sec ax + i \cos ax] dx =$$

$$= e^{iax} \left[\frac{-\cos ax}{a} - \frac{i}{a} \log(\sec ax + \tan ax) + \frac{i \sin ax}{a} \right]$$

$$= \frac{-e^{iax}}{a} [(\cos ax - i \sin ax) + i \log(\sec ax + \tan ax)]$$

$$= \frac{-e^{iax}}{a} [e^{-iax} + i \log(\sec ax + \tan ax)]$$

$$= -\frac{1}{a} [1 + i e^{iax} \log(\sec ax + \tan ax)] \quad \text{--- (1)}$$

Changing i to $-i$ in eqⁿ. (1),

$$\frac{1}{D+ia} \tan ax = -\frac{1}{a} [1 - i e^{-iax} \log(\sec ax + \tan ax)]$$

$$\therefore \text{P.I.} = \frac{1}{2ia} \left\{ \frac{-1}{a} [1 + i e^{iax} \log(\sec ax + \tan ax)] + \frac{1}{a} [1 - i e^{-iax} \log(\sec ax + \tan ax)] \right\}$$

$$= \frac{1}{2ia^2} \left[-2i \left(\frac{e^{iax} + e^{-iax}}{2} \right) \log(\sec ax + \tan ax) \right]$$

$$= \frac{1}{2ia^2} \left[-2i \cos ax \cdot \log(\sec ax + \tan ax) \right]$$

$$= -\frac{1}{a^2} \cdot \cos ax \cdot \log (\sec ax + \tan ax)$$

Hence, the C.S. is,

$$y = C.F. + P.I.$$

$$= C_1 \cos ax + C_2 \sin ax - \frac{1}{a^2} \cos ax \log (\sec ax + \tan ax) \quad \underline{\text{Ans}}$$

33. $[D^4 + 2D^2 + 1]y = x^2 \cos x$

Solⁿ:

The A.E. is,

$$m^4 + 2m^2 + 1 = 0$$

$$\Rightarrow (m^2 + 1)^2 = 0$$

$$\Rightarrow (m^2 + 1)(m^2 + 1) = 0$$

$$\Rightarrow m^2 = -1, m^2 = -1$$

$$\Rightarrow m^2 = i^2, m^2 = i^2$$

$$\Rightarrow m = \pm i, \pm i$$

$$\begin{aligned} C.F. &= e^{0 \cdot x} [(C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x] \\ &= (C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x \end{aligned}$$

$$P.I. = \frac{1}{(D^4 + 2D^2 + 1)} \cdot x^2 \cos x$$

$$= R.P. \frac{1}{(D^4 + 2D^2 + 1)} x^2 \cdot e^{ix}$$

$$= R.P. \frac{e^{ix}}{[(D+i)^2+1]^2} \cdot x^2$$

$$= R.P. \frac{e^{ix}}{\{(D^2+2iD+i^2)+1\}^2} \cdot x^2$$

$$= R.P. \frac{e^{ix}}{(D^2+2iD-1+1)^2} \cdot x^2$$

$$= R.P. \frac{e^{ix}}{(D^2+2iD)^2} \cdot x^2$$

$$= R.P. \frac{e^{ix} x^2}{4i^2 D^2 \left\{ 1 + \frac{D^2}{2iD} \right\}^2}$$

$$= R.P. \frac{-e^{ix}}{4D^2} \left\{ 1 + \frac{D}{2i} \right\}^{-2} x^2$$

$$= R.P. \frac{-e^{ix}}{4D^2} \left\{ 1 - 2 \cdot \frac{D}{2i} + 3 \cdot \left(\frac{D}{2i} \right)^2 - \dots \right\} x^2$$

$$= R.P. \frac{-e^{ix}}{4D^2} \left\{ 1 + Di - \frac{3}{4} D^2 + \dots \right\} x^2$$

Neglecting higher order of D ,

$$R.P. \frac{-e^{ix}}{4D^2} \left\{ x^2 + iD(x^2) - \frac{3}{4} D^2(x^2) \right\}$$

$$= \text{R.P.} \frac{-e^{ix}}{4D^2} \left\{ x^2 + 2xi - \frac{3}{4} x^2 \right\}$$

$$= \text{R.P.} \frac{-e^{ix}}{4D^2} \left\{ x^2 + 2xi - \frac{3}{2} \right\}$$

$$= \text{R.P.} \frac{-1}{4D^2} [\cos x + i \sin x] [x^2 + 2xi - 3/2]$$

1st Method
Long Process

$$= \text{R.P.} \frac{-1}{4D^2} \left\{ x^2 \cos x + i x^2 \sin x \right\}$$

$$= \text{R.P.} e^{ix} \left\{ \frac{-1}{4D} \int \left(x^2 + 2xi - \frac{3}{2} \right) dx \right\}$$

$$= \text{R.P.} e^{ix} \left\{ \frac{-1}{4} \int \left(\frac{x^3}{3} + \frac{2x^2}{2} i - \frac{3}{2} x \right) dx \right\}$$

$$= \text{R.P.} e^{ix} \left\{ \frac{-1}{4} \left(\frac{x^4}{12} + \frac{x^3}{3} i - \frac{3x^2}{2 \times 2} \right) \right\}$$

$$= \text{R.P.} e^{ix} \left\{ \frac{-x^4}{48} - \frac{x^3}{12} i + \frac{3x^2}{16} \right\}$$

$$= \text{R.P.} (\cos x + i \sin x) \left[\frac{-x^4}{48} - \frac{x^3}{12} i + \frac{3x^2}{16} \right]$$

$$= \text{R.P.} \left\{ \frac{-x^4}{48} \cos x - \frac{ix^3}{12} \cos x + \frac{3x^2}{16} \cos x \right.$$

$$\left. - \frac{ix^4}{48} \sin x - \frac{i^2 x^3}{12} \sin x + \frac{i 3x^2}{16} \sin x \right\}$$

$$= \int \left\{ \frac{-x^4}{48} \cos x + \frac{3x^2}{4} \cos x + \frac{x^3}{12} \sin x \right\}$$

Hence, the C.S. is,

$$y = C.F. + P.I.$$

$$= (C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x - \frac{x^4}{48} \cos x + \frac{3x^2}{4} \cos x + \frac{x^3}{12} \sin x \quad \underline{\underline{Ans}}$$

$$34. [D^4 - 1]y = x \sin x$$

Solⁿ:

The A.E. is,

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m^2 = 1, m^2 = -1$$

$$\Rightarrow m = \pm 1, m = \pm i$$

$$C.F. = C_1 e^{-x} + C_2 e^x + C_3 \cos x + C_4 \sin x$$

$$P.I. = \frac{1}{(D^4 - 1)} \cdot x \sin x$$

$$= I.P. \frac{1}{(D^4 - 1)} \cdot x e^{ix}$$

$$= I.P. \frac{1}{(D^2 - 1)(D^2 + 1)} \cdot x \cdot e^{ix}$$

$$= I.P. e^{ix} \frac{1}{[(D+i)^2 - 1][(D+i)^2 + 1]} \cdot x$$

$$= I.P. e^{ix} \frac{1}{(D^2 + 2iD - 2)(D^2 + 2iD)} \cdot x$$

$$= I.P. e^{ix} \frac{1}{\{D^4 + 2iD^3 - 2D^2 + 2iD^3 - 4D^2 - 4iD\}} \cdot x$$

$$= \text{I.P. } e^{ix} \frac{1}{\{D^4 + 4iD^3 - 6D^2 - 4iD\}} x$$

$$= \text{I.P. } e^{ix} \frac{1}{-4iD \left\{ \frac{-D^4}{4iD} - \frac{4iD^3}{4iD} + \frac{6D^2}{4iD} + 1 \right\}} x$$

$$= \text{I.P. } \frac{e^{ix}}{-4iD} \frac{1}{\left\{ 1 - \left[\frac{D^3}{4i} + D^2 + \frac{i3D}{2} \right] \right\}} x$$

$$= \text{I.P. } \frac{-e^{ix}}{4iD} \left[1 - \left(\frac{-iD^3}{4} + D^2 + i\frac{3D}{2} \right) \right]^{-1} x$$

$$= \text{I.P. } \frac{ie^{ix}}{4D} \left[1 + \left(\frac{-iD^3}{4} + D^2 + i\frac{3D}{2} \right) + \dots \right] x$$

Removing higher order of D ,

$$= \text{I.P. } \frac{ie^{ix}}{4D} \left[1 + \left(+i\frac{3D}{2} \right) \right] x$$

$$= \text{I.P. } \frac{ie^{ix}}{4D} \left(x + i\frac{3}{2} D(x) \right)$$

$$= \text{I.P.} \frac{e^{ix}}{4D} \left(ix - \frac{3}{2} \right)$$

$$= \text{I.P.} \frac{e^{ix}}{4D} \int \left(ix - \frac{3}{2} \right) dx$$

$$= \text{I.P.} \frac{e^{ix}}{4} \left(\frac{ix^2}{2} - \frac{3x}{2} \right)$$

$$= \text{I.P.} \left(\frac{ix^2}{8} - \frac{3x}{8} \right) (\cos x + i \sin x)$$

$$= \text{I.P.} \left[\frac{ix^2}{8} \cos x - \frac{3x}{8} \cos x + i^2 \frac{x^2}{8} \sin x - i \frac{3x}{8} \sin x \right]$$

$$= \frac{x^2}{8} \cos x - \frac{3x}{8} \sin x$$

Hence, the C.S. is,

$$y = \text{C.F.} + \text{P.I.}$$

$$= C_1 e^{-x} + C_2 e^x + C_3 \cos x + C_4 \sin x + \frac{x^2}{8} \cos x - \frac{3x}{8} \sin x$$